

CPE 381 CLASSWORK 03: MATHEMATICAL PRELIMINARIES

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes

Due: August 31, 2026 (In-Class)

1. Solve the differential equation $ty' + 3y = 0$, given $y(1) = 2$, assuming $t > 0$. Here $y' = \frac{dy}{dt}$. **Hint 1:** you will need to rearrange terms so that derivative term is on the left hand side and non-derivative terms on the right hand side, and then integrate. **Hint 2:** $\int \frac{1}{t} dt = \ln(t) + C$. **(5 Points)**

Answer

2. Find the derivate of $f(x) = x^2$ using the first priciple of the derivate. **(5 Points)**

Answer

3. Find the second, fourth, seventh, and ninth terms of the geometric progression with first term $a = 2$ and ratio $r = 3$. Does the sum of this geometric progression converge? (4 + 1 Points)

Answer

4. Find area under the curve for the shaded region in Figure 1. **Hint:** $\int x^n dx = \frac{1}{n+1}x^{n+1}$. (5 Points)

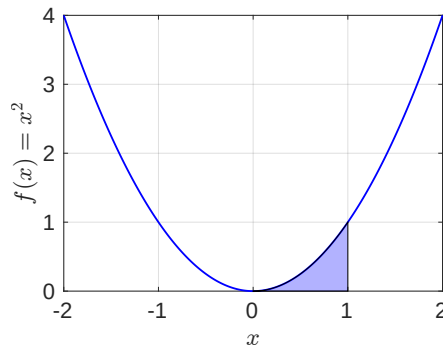


Figure 1: Area under curve

Answer